SET 2430 ▯ Assignment 2: Python OOP & Foundations

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Python Fundamentals, OOP, and Variable Arguments

Part A. Prime Number Check

Write a function is_prime(n) that:

1. Validates strict integer input (reject floats, strings, booleans, and None).

2. Numbers less than or equal to 1 are not prime.

3. Check factors efficiently up to int(n**0.5) + 1.

Part B. Remove Adjacent Duplicates

Write a function remove_adjacent_duplicates(id_list) that:

1. Iterates through the list using a stack-based list pattern.

2. If the current element matches the top/last element, pop it.

3. Otherwise, append the element. Return the resulting deduplicated list.

Part C. Display Student Information Using Variable Arguments

Write a function disStuInfo(schoolID, *last_names, **first_name_email) that:

1. Synchronizes positional *last_names with keyword **first_name_email.

2. Prints schoolID, last name, first name, and email for matching pairs.

3. If unmatched, outputs 'unmatched' and prints the remaining names.

Part D. Staff OOP Class Implementation

Create a Staff class with:

1. Instance variables: name, _position, pay.

2. @property getter and @position.setter with validation ('Manager', 'Intern', 'developer').

3. calPay() method calculating hours * hourly_rate.